SREYA REDDY CHINTHALA

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Education

University of Southern California — Los Angeles, CA
Master of Science in Computer Science (GPA: 3.76) — *Aug. 2023 – May 2025*

Christ University — Bangalore, India
Bachelor of Technology in Computer Science and Engineering (GPA: 3.95) — *May 2018 – Jul. 2022*

Skills

Languages: Python, Java, SQL, JavaScript, HTML/CSS, SwiftUI
Automation & Testing: Python (pytest, Playwright, WebKit), Test Automation, API Testing, Regression Testing, Integration Testing, Accessibility Testing (axe-core), Data Validation, Root Cause Analysis, Bug Isolation & Reporting
Backend & Tools: REST APIs, Flask, Node.js, ReactJS, Docker, Git, GitHub Actions, CI/CD Pipelines
Database and Cloud: PostgreSQL, MongoDB, Relational & NoSQL Databases, AWS (Lambda, S3, EC2, DynamoDB)
Machine Learning: TensorFlow, Keras, PyTorch, OpenCV, Scikit-Learn, Numpy, Pandas

Experience

Software Engineer — Los Angeles, CA
Prime Healthcare — *Nov. 2025 – Present*

- Engineered 6+ automation workflows in Python and JavaScript with data validation and error handling; validated across multiple test scenarios and edge cases, cutting manual effort by **5+ hours/week**
- Analyzed production logs and application data to identify root causes of critical failures, validated fixes through **regression testing**, and deployed releases through CI/CD pipelines across multiple environments

Software Engineer Intern — Torrance, CA
Easley Dunn Productions — *Sep. 2025 – Oct. 2025*

- Built Python automation pipelines to extract and validate **VR system logs**, reducing manual review by **70%**
- Developed REST API integrations with JSON Schema validation and error handling to aggregate metrics and power a **real-time analytics dashboard**

Machine Learning Intern — Los Angeles, CA
Neuroimaging and Informatics Institute, USC — *Mar. 2024 – Jul. 2024*

- Built and validated modular ML pipelines in Python and PyTorch for MRI preprocessing; designed **evaluation frameworks** to benchmark model outputs and catch regressions across experimental iterations
- Automated model training and evaluation workflows via CI/CD pipelines, improving reproducibility and reducing experiment iteration time by **30%** across research teams

Machine Learning Engineer — Daegu, South Korea
Vision and Intelligent Systems Lab — *Apr. 2022 – Jul. 2023*

- Led **comparative evaluation** of **25+ AI models** across 7 architectures on 2,815 medical videos, designing test frameworks to benchmark accuracy, F1, Cohen-Kappa, and computational efficiency
- Investigated **model failure modes** by analyzing false positives and negatives against physician-annotated ground truth; identified edge cases in imbalanced datasets and validated fixes across experimental iterations
- Delivered **90.6% accuracy** and **89.1% F1-score** with 40% faster inference; findings published in IEEE Access

Machine Learning Intern — Daegu, South Korea
Intelligent Signal Processing Lab — *Jan. 2021 – Oct. 2021*

- Designed and validated a foreground extraction pipeline in OpenCV to improve AI model accuracy by 3.33%; benchmarked against baseline across 8,096 images and published findings in IEEE

Projects

Apple.com Automation Framework | *Python, Playwright, WebKit, pytest, axe-core*

- Built a cross-browser test automation framework for Apple.com using WebKit and Chromium with Page Object Model, 28 tests covering navigation, search, accessibility (WCAG), and performance; filed 2 bugs to Apple (**FB22823503, FB22851540**); daily CI via GitHub Actions with HTML reports (GitHub)

WebRTC Reliability Framework | *Python, JavaScript, Playwright, WebKit, pytest*

- Developing an automated test framework using Playwright and injected JavaScript to benchmark **WebRTC peer connection** reliability, latency, and network degradation across WebKit and Chromium *(in progress)* (GitHub)

AI Research Agent (RAG System) | *Python, LangChain, RESTful API*

- Designed an agentic RAG pipeline with OpenAI, Anthropic, and Google GenAI with prompt templates and Pydantic parsers, integrating web search, Wikipedia, and custom tools to automate research reports (GitHub)

iOS App – Museum Artifact Exploration | *SwiftUI, CoreML, ARKit, RealityKit, Maps SDK*

- Developed an interactive app using CoreML for **on-device inference** for artifact recognition from camera capture or gallery upload, displaying details, AR visualizations, and museum location via Google Maps (GitHub) (Demo)

Audio-Textual Depression Detection with Multi-modal Transformer | *Python, PyTorch*

- Developed a multi-modal Transformer integrating BERT text embeddings with NetVLAD audio representations, achieving **75.8% accuracy** and **0.81 F1**, with **attention-based** interpretability (GitHub)